

# **Process Characterization Report**

**A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCR-004**

Process Development / Manufacturing Science & Technology (MSAT)

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## Approvals

Table 1: Approval record (synthetic; signatures not applied).

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| Approver | Quality Assurance | —                | —         | —    |

## Abbreviations

AMV, analytical method validation; BLA, Biologics License Application; Cpk, process capability index; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; NTU, nephelometric turbidity unit; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; rcf, relative centrifugal force; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

## Executive summary

Harvest and clarification is the primary recovery step of the A-Mab drug-substance process. It removes cells and cell debris from the production bioreactor broth by continuous disk-stack centrifugation (SOP-2005), and it clarifies the centrate through a depth filter train and a sterile filter (SOP-2006). The step forms no product-quality attribute and is credited with no impurity clearance. What it delivers is product recovery and a consistent, low-solids feed to Protein A capture (PCR-005).

The characterization was executed under PCP-004 in a qualified scale-down model, and it follows the Stage 1 process-design expectations of the lifecycle approach (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 (International Council for Harmonisation 2009), with the A-Mab case study as the process reference throughout (CMC Biotech Working Group 2009). Three parameters were characterized, each one at a time over a range two to three times the width of its normal operating range. Centrifugation and depth-filter loading are key process parameters and post-clarification turbidity is a general process parameter, and the step carries no critical process parameter, because none of its parameters is linked to a critical quality attribute.

In the nominal commercial-scale simulation the step recovered 97.7% of the product presented to it, which is 64.1 kg of the 65.6 kg leaving the bioreactor, and it accounts for 13.8% of the total product loss across the drug-substance train. Post-clarification turbidity at the set-point condition was 5 NTU, in the middle of its normal operating range of 1–10 NTU. Both results were obtained over the characterized ranges given in §7 and in a scale-down model, so they are bounded by that model and by those ranges.

The clarified harvest carries about 500,000 ng/mg host cell protein (HCP) into capture. That burden is set by the culture and cleared by the purification train, not by this step. Protein A capture alone reduces it 55-fold (PCR-005), and the drug substance meets its limit of 100 ng/mg with a capability of 6.14 (§8). Two deviations were recorded, investigated and closed. Both were retained without altering a parameter classification or a characterized range (§13). The step is ready for Stage 2 with the ranges given in §7, and its outcome rolls up into PCMR-001.

# 1 Introduction

## 1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity, and it is produced by fed-batch mammalian cell culture and purified by a platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration or diafiltration. The train and the role of each step are given in Table 1.1. Harvest and clarification sits between the production bioreactor (PCR-003) and capture, and it is the only step of the train that handles unclarified broth.

Table 1.1: The A-Mab drug-substance process train and the principal role of each step.

| Step | Unit operation                   | Principal role                                                               |
|------|----------------------------------|------------------------------------------------------------------------------|
| 3    | Production Bioreactor            | Forms the glycan, charge-variant and aggregate CQAs (design-space step)      |
| 4    | Harvest and Clarification        | Primary recovery and clarification; forms no product-quality CQA             |
| 5    | Protein A Chromatography         | Capture; sets leached Protein A; principal HCP and DNA clearance             |
| 6    | Low-pH Viral Inactivation        | Low-pH hold; sets the cumulative XMuLV (enveloped) clearance                 |
| 7    | Cation Exchange Chromatography   | Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance    |
| 8    | Anion Exchange Chromatography    | Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA |
| 9    | Small-Virus Retentive Filtration | Dedicated small-virus removal; principal MVM clearance                       |
| 10   | Ultrafiltration / Diafiltration  | Formulation / mass balance; delivers drug substance at target concentration  |

The step operates in two stages. In the first, the whole broth is fed continuously to a disk-stack centrifuge, which sediments cells and the larger debris and discharges them as a solids stream. The centrate is then passed through a depth filter train, which removes the fine solids that the centrifuge cannot sediment, and finally through a sterile filter into the capture feed vessel, where solids removal is judged by the turbidity of the clarified harvest measured by nephelometry.

The step forms no critical quality attribute and reduces none, because it has no chromatographic, chemical or size-based mechanism that acts on the antibody, so the

glycan, charge-variant and aggregate attributes formed in the bioreactor (PCR-003) pass through it unchanged. Host cell protein and residual DNA enter it with the broth and leave it with the clarified harvest. Their clearance is the work of the purification steps, and the credit for it is given in PCR-005, PCR-007 and PCR-008.

## 1.2 Objectives

The objectives of this characterization were:

- (i) To confirm that the step forms and modifies no critical quality attribute over the ranges studied.
- (ii) To quantify product recovery and its variability at commercial-equivalent scale.
- (iii) To establish the ranges of centrifugation force and depth-filter loading over which clarification remains acceptable.
- (iv) To classify each parameter of the step under SOP-4001.
- (v) To define the in-process controls that assure a consistent feed to Protein A capture.

## 1.3 Regulatory and scientific basis

The study is a Stage 1 process-design activity in the lifecycle sense (U.S. Food and Drug Administration 2011), and it is reported here as part of the record supporting the Biologics License Application. The development approach follows ICH Q8 and ICH Q11, and criticality is treated as a continuum under ICH Q9 (International Council for Harmonisation 2009, 2023b, 2012). Attributes are not sorted into critical and non-critical alone. They carry a level of criticality that reflects both the impact of the attribute and the uncertainty in that impact, and the parameters of a step are classified against the attributes they affect.

No viral-clearance claim is made for this step, so ICH Q5A is not applied here (International Council for Harmonisation 2023a), and the cumulative viral-clearance claim for the drug substance rests on low-pH inactivation (PCR-006), anion exchange (PCR-008) and small-virus retentive filtration (PCR-009), where it is consolidated into the summary held in PCMR-001.

## 1.4 Related documents

The risk basis for the scope of this study is RA-001, the paired protocol is PCP-004, the transfer context is PTP-001, and the cross-references are listed in Table [1.2](#).

Table 1.2: Related documents in the A-Mab characterization package.

| Docu-<br>ment ID | Title                                                               | Relationship                        |
|------------------|---------------------------------------------------------------------|-------------------------------------|
| PTP-001          | Process Transfer Plan (A-Mab Drug Substance)                        | Parent — transfer scope             |
| RA-001           | Pre-Characterization Process Risk Assessment (A-Mab Drug Substance) | Parent — risk basis for study scope |
| PCMP-001         | Process Characterization Master Plan (A-Mab Drug Substance)         | Parent — master plan                |
| PCP-004          | Process Characterization Plan (Harvest and Clarification (Step 4))  | Sibling — paired document           |
| PCMR-001         | Process Characterization Master Report (A-Mab Drug Substance)       | Rolls up into                       |

## 2 Prior knowledge and quality risk basis

### 2.1 Platform and prior-product knowledge

Centrifugation followed by depth filtration is the established primary-recovery platform at Novacyte Biologics, and it has been used without change for three previous humanized IgG1 products (X-Mab, Y-Mab and Z-Mab). The mechanisms are physical and well understood. Solids are removed in the centrifuge by sedimentation, so removal improves as relative centrifugal force (rcf) rises and as feed flow falls. The depth filter removes the fine solids that remain, by a combination of size exclusion and adsorption, and its capacity is expressed as the volume filtered per unit of filter area ( $L/m^2$ ). As that capacity is consumed, retained solids begin to pass and the turbidity of the filtrate rises.

Two expectations follow from these mechanisms, and the study was designed to test them. The first is that clarification performance depends on centrifugation force and on filter loading, and that the loading limit is reached before the sedimentation limit at the platform set-points. The second is that neither setting changes product quality, because neither has a mechanism that acts on the antibody, and prior product experience supports both expectations, as does the treatment of primary recovery in the A-Mab case study (CMC Biotech Working Group 2009). That experience reduced the scope of this study to a univariate assessment of three parameters.

The impurity burden entering capture is set upstream. Host cell protein and DNA are released by the culture, and the burden rises with culture duration and with the fall in end-of-culture viability that accompanies it. In the nominal batch the culture was harvested at a viability of 67.1 % and a titre of 4.37 g/L. Clarification carries that burden forward, and it neither adds to it nor reduces it.

### 2.2 Quality attributes in scope

This step sets no critical quality attribute. The attributes it carries forward, and against which its parameters were assessed, are given in Table 2.1. Host cell protein is the most consequential of the three, at moderate to high criticality, because it is the impurity present in the largest amount in the clarified harvest. Aggregate is of high criticality and is the one attribute with a plausible mechanism at this step, since shear at the centrifuge feed zone could in principle raise it. Residual DNA is of very low criticality on its own, and is cleared by a large margin downstream.

Table 2.1: Quality attributes carried forward by the step, with acceptance criteria and criticality. None is formed or cleared here.

| CQA                     | Category                 | Acceptance      | Criticality | Tool #1 |
|-------------------------|--------------------------|-----------------|-------------|---------|
| Host Cell Protein (HCP) | process                  | 0-100 ng/mg     | M-H         | 36      |
| Aggregates (HMW)        | impurity<br>size variant | 0-5 % HMW (SEC) | H           | 60      |
| Residual DNA            | process<br>impurity      | 0-0.001 ng/dose | VL          | 6       |

## 2.3 Risk-based prioritization and parameter selection

RA-001 ranked every parameter of the step for potential impact on quality attributes and for potential interaction with other parameters, and all three fell into the band for which a univariate assessment is sufficient, with none reaching the band that warrants a multivariate design. The reason is the same in each case. No parameter of the step has a mechanism by which it can act on a quality attribute, so the risk it carries is a process-performance risk and not a quality risk.

The characterized ranges were widened beyond the routine control ranges so that the study would describe process robustness and not only the target condition. Centrifugation was studied over 6000-12000 g, which is 3 times the width of its normal operating range, and depth-filter loading over 80-160 L/m<sup>2</sup> and turbidity over 1-20 NTU, each about 2 times the width of the corresponding normal operating range. The lower centrifugation edge was set where prior platform data show that sedimentation becomes incomplete, and the upper depth-filter edge was set at the loading beyond which the platform filter is no longer guaranteed by its vendor.

## **3 Materials and methods**

### **3.1 Scale-down model and its qualification**

A scale-down model of the harvest and clarification step was qualified against commercial-scale data before the study, under SOP-1001. The centrifuge model is a bench disk-stack machine operated at the same relative centrifugal force and at a feed flow scaled to hold the equivalent settling area constant (the sigma factor), so that the sedimentation duty of the commercial machine (SOP-2005) is reproduced. The depth filter train uses the same filter media as the commercial train and is operated at the same loading in litres per square metre, so that capacity translates directly.

Qualification compared step yield and post-clarification turbidity between the model and the commercial-scale process at the target condition, and found no difference that the analytical methods could resolve. The qualification record is held under SOP-1001 and is not reproduced here. Two limits of the model should be read together with its results. It reproduces the sedimentation and capacity duties, but it does not reproduce the full shear history of a commercial disk-stack machine, and it operates at hold times shorter than those of a commercial harvest. Both limits are addressed by confirming yield and turbidity at scale during process performance qualification.

### **3.2 Operation**

The step was operated under SOP-2005 for centrifugation and SOP-2006 for depth and sterile filtration, harvest begins at the end of the production culture, and the whole broth is fed to the centrifuge without a hold. The centrate is clarified immediately, and the clarified harvest is held in the capture feed vessel until Protein A processing begins. Filter media are flushed and integrity-tested before use under SOP-2006.

Inputs that are controlled to platform specification were not characterized here. Filter media, buffers and the sterile filter are qualified by identity and quality on receipt, and their control is exercised through the material specifications and not through a characterization range.

### **3.3 Analytical methods**

Turbidity is the primary in-process measure of clarification and was determined by nephelometry (AMV-3015), while host cell protein was measured by ELISA (AMV-3012)

and residual DNA by qPCR (AMV-3014), both to describe the burden handed to capture. Aggregate was followed by SEC-HPLC (AMV-3011) to confirm that clarification does not change the size distribution. The controlled documents that govern the step are listed in Table 3.1, and method performance is given in Appendix B.

Table 3.1: Procedures and validated analytical methods applied to the step.

| Reference | Title                                                            | Type              |
|-----------|------------------------------------------------------------------|-------------------|
| SOP-1001  | Qualification of Scale-Down Models                               | SOP               |
| SOP-2005  | Harvest by Continuous Disk-Stack Centrifugation                  | SOP               |
| SOP-2006  | Clarification by Depth and Sterile Filtration                    | SOP               |
| SOP-4001  | Process-Parameter Classification and Control-Strategy Definition | SOP               |
| AMV-3015  | Turbidity by Nephelometry (NTU)                                  | Method validation |
| AMV-3012  | Host-Cell Protein ELISA                                          | Method validation |
| AMV-3014  | Residual DNA (qPCR)                                              | Method validation |
| AMV-3011  | Size-Variants (SEC-HPLC)                                         | Method validation |

### 3.4 Sampling plan

Each run was sampled at three points: the whole broth at the start of harvest, the centrate once the centrifuge had reached steady discharge, and the clarified harvest at the end of filtration. Turbidity was measured in-process on the clarified harvest and the remaining assays were run on the end-of-filtration sample, and retained samples of every clarified harvest are held for the campaign, so that a later question about a run can be answered on retained material.

### 3.5 Statistical methods

Each parameter was varied one at a time across its characterized range with the other parameters held at their set-points. Responses were summarized descriptively, and no multivariate model was fitted. That choice follows from the risk assessment. A response surface is the predictive model that defines a design space, and it is worth building only where a parameter acts on a quality attribute, which at this step is true of none of them, so there is no response to model and no design space to define.

Commercial-scale capability for the attributes this step hands forward was taken from the Monte-Carlo simulation of 2,000 batches that supports the whole train, and it is reported in §8 as cross-step context. Capability is expressed as a one-sided Cpk against the acceptance limit.

## 4 Study design

### 4.1 Parameters, ranges and the knowledge space

The parameters, their set-points, their normal operating ranges and the ranges over which they were characterized are given in Table 4.1, together with the final classification and the study type. The characterized range is the knowledge space of the step. It is wider than the normal operating range on both sides, and the normal operating range is nested inside it with margin at each edge.

Table 4.1: Parameters of the harvest and clarification step, with set-points, normal operating ranges, characterized ranges, final classification and study type.

| Parameter                    | Unit             | Set-point | NOR        | Char.<br>range | Class | Study      |
|------------------------------|------------------|-----------|------------|----------------|-------|------------|
| Centrifugation (rcf)         | g                | 9,000     | 8000-10000 | 6000-12000     | KPP   | univariate |
| Depth filter load            | L/m <sup>2</sup> | 120       | 100-140    | 80-160         | KPP   | univariate |
| Post-clarification turbidity | NTU              | 5         | 1-10       | 1-20           | GPP   | univariate |

Post-clarification turbidity appears in the register as a parameter, but it behaves as an outcome and is not set by the operator. It is the measured result of the centrifugation and filtration settings and the attribute by which the clarified harvest is accepted for capture, and it is characterized and controlled as a range for that reason.

### 4.2 Univariate assessment

No designed experiment was executed at this step, and none is reported. Centrifugation force was assessed across 6000-12000 g with depth-filter loading at its set-point of 120 L/m<sup>2</sup>. Depth-filter loading was then assessed across 80-160 L/m<sup>2</sup> with centrifugation at its set-point of 9000 g, and turbidity was recorded on every run, with product quality measured on the clarified harvest of each. Verification runs at the set-point condition bracketed the study, so that a drift in the model or in the assays over the campaign would have been visible.

The limitation of a univariate design is that it cannot resolve an interaction between the two settings. That limitation is accepted here because the two act on different

solids populations, the centrifuge on the sedimentable fraction and the filter on the fine fraction, and because neither acts on product quality. An interaction between them could change clarification performance, and it could not change a quality attribute.

## 5 Results

### 5.1 Step yield and mass balance

The step recovers almost all of the product presented to it. In the nominal commercial-scale simulation 65.6 kg of A-Mab entered the centrifuge and 64.1 kg left the sterile filter, a step yield of 97.7%. The loss of 1.5 kg is product held in the discharged solids and in the wetted filter train. Step yield varies from batch to batch with a coefficient of variation of 2.0%.

Recovery showed no dependence on centrifugation force or on depth-filter loading over the ranges assessed, and the loss is dominated by hold-up in the solids and in the filter media, which does not change appreciably across those ranges. The step accounts for 13.8% of the total product loss across the train, which delivers a cumulative yield of 83.2% (Figure 5.1), and yield at this step is therefore a process-performance concern and not a constraint on the process.

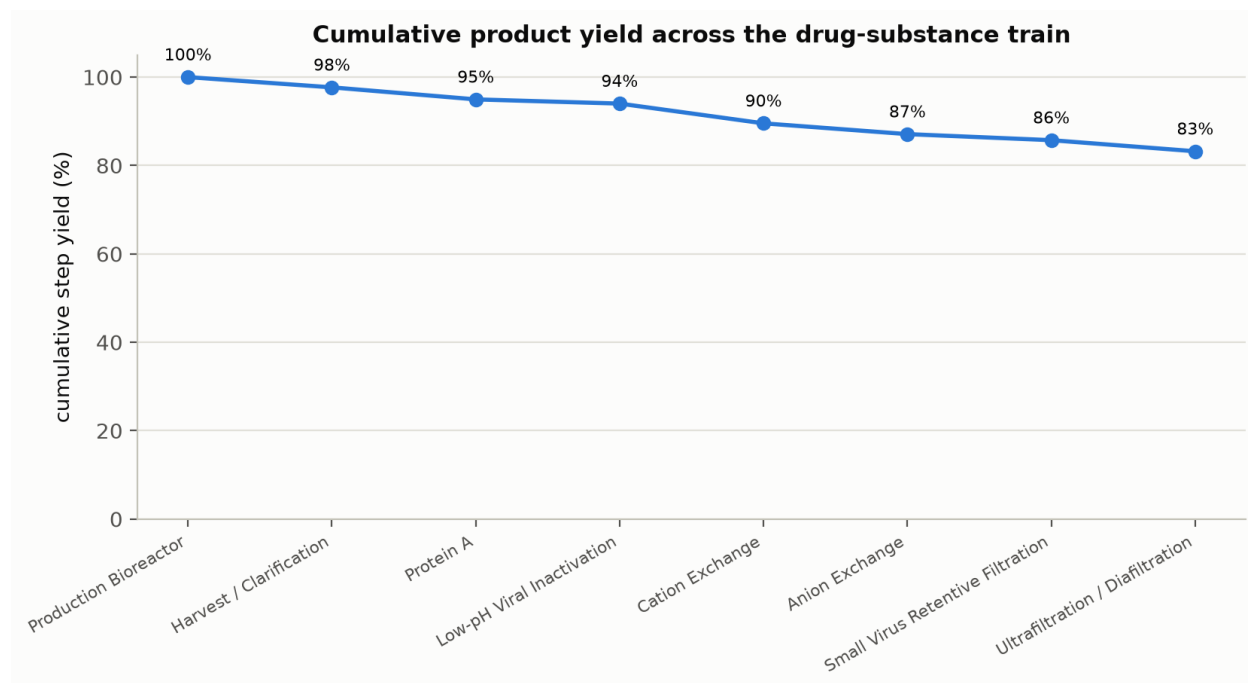


Figure 5.1: Cumulative product yield across the drug-substance train, with harvest and clarification as the first loss in the sequence.

## 5.2 Clarification performance and filter capacity

Clarification met its in-process expectation across the ranges studied, with one excursion that is described below. Post-clarification turbidity at the set-point condition was 5 NTU, in the middle of the normal operating range of 1–10 NTU. The nephelometric method has a precision of 4 % and a limit of quantitation of 0.5 NTU (Appendix B), so the excursion described below is well outside assay variability.

Turbidity rose as depth-filter loading approached the top of its characterized range, which is the expected consequence of consuming filter capacity and is the mechanism set out in §2.1. One run exceeded the upper limit of the normal operating range and reached 12.5 NTU, which is 2.5 NTU above that limit, at a loading near the maximum characterized value of 160 L/m<sup>2</sup> (DEV-004-02, §13). The value remains inside the characterized turbidity range of 1–20 NTU, and the run was retained. Centrifugation force produced no comparable trend over 6000–12000 g, which is consistent with sedimentation being complete well below the upper edge of that range.

## 5.3 Impurity burden carried to Protein A capture

The clarified harvest carries a large host cell protein burden into capture. At the nominal culture condition the level is about 500,000 ng/mg by ELISA, and residual DNA is present at a correspondingly high level. Neither is reduced here. Both are carried forward to Protein A capture, which is the first step of the train with a mechanism that separates them from the antibody (affinity binding).

Clearance is a property of the purification train and not of this step. Protein A capture reduces host cell protein 55-fold, to 9,125 ng/mg in the capture pool (PCR-005). Cation exchange and anion exchange then complete the reduction (PCR-007, PCR-008), and the drug substance meets its host cell protein limit of 100 ng/mg with a mean of 15.2 ng/mg (§8). The contribution of harvest and clarification to that outcome is the consistency of the material it hands over, not a log of clearance.

## 5.4 Product quality across the step

No quality attribute changed across clarification, and aggregate, charge-variant and glycan levels measured in the clarified harvest were the same as those in the bioreactor broth within the variability of the methods, which is what a step with no mechanism to act on the antibody should give. Aggregate was followed most closely, by SEC-HPLC, because shear at the centrifuge feed zone is the one plausible product-quality risk here. No increase was seen over the centrifugation range assessed.

That null result is a finding, and it is retained in the knowledge space. It is the evidence that supports classifying both settable parameters as key and not critical (§9), and it is the reason the step contributes nothing to the design space of the drug substance (§6).

The result is bounded by the ranges in Table [4.1](#), by the scale-down model described in §3.1, and by the single culture condition from which the harvest material was taken.

## 6 Design space

This step contributes no design space, since a design space is the multivariate region of parameter settings that is known to give acceptable values of the critical quality attributes, and it can only be defined for a step whose parameters act on such an attribute. Harvest and clarification has no parameter that does, so the region is empty and its parameters are controlled as ranges instead, and the A-Mab case study reaches the same conclusion for the steps of the process that do not affect product quality (CMC Biotech Working Group 2009).

Two consequences follow. Movement of centrifugation force or depth-filter loading within the ranges given in §7 is not a change to the design space of the drug substance, and it is managed under the quality system as a process-performance matter (International Council for Harmonisation 2008). Operation outside those ranges is a change, and it requires the assessment set out in the control strategy, since the ranges themselves are bounded by the scale-down model and by the single culture condition studied.

## 7 Proven acceptable ranges

Each parameter was assessed univariately over its characterized range, and each range is proven acceptable over its full width for the process-performance criteria of the step. The ranges are given in Table 7.1. The acceptance basis here is not a drug-substance specification, because the step governs no attribute that carries one, and it is instead the pair of in-process criteria that the step must meet, which are the recovery of the product and the delivery of a clarified harvest fit for capture.

Table 7.1: Proven acceptable ranges for the harvest and clarification parameters. The proven acceptable range is the univariately characterized range in each case.

| Parameter                    | Unit             | Set-point | NOR        | Proven acceptable range | Class |
|------------------------------|------------------|-----------|------------|-------------------------|-------|
| Centrifugation (rcf)         | g                | 9,000     | 8000-10000 | 6000-12000              | KPP   |
| Depth filter load            | L/m <sup>2</sup> | 120       | 100-140    | 80-160                  | KPP   |
| Post-clarification turbidity | NTU              | 5         | 1-10       | 1-20                    | GPP   |

Centrifugation is proven acceptable over 6000-12000 g and depth-filter loading over 80-160 L/m<sup>2</sup>, and the normal operating ranges of 8000-10000 g and 100-140 L/m<sup>2</sup> sit inside those ranges with margin at both edges. Turbidity is proven over 1-20 NTU in the sense that clarified harvest across that range was acceptable to capture, but it is controlled to 1-10 NTU because the upper part of the wider range is only reached when filter capacity is nearly exhausted. The worst case that the ranges must satisfy is therefore the high-loading corner, where filter capacity is most nearly consumed and turbidity is highest.

These are univariate ranges and not a multivariate region. They were established with the other parameters at their set-points, so they carry the limitation set out in §4.2. They are bounded by the scale-down model, by the ranges studied, and by the culture condition from which the harvest material came.

## 8 Process capability and robustness

The step has no process capability of its own to report, because it governs no quality attribute and therefore has no acceptance limit to be capable against. The capability that its output eventually supports is given in Table 8.1 for the two impurities it hands forward, and both figures are commercial-scale results from the Monte-Carlo simulation of 2,000 batches that belongs to the purification train.

Table 8.1: Commercial-scale capability for the impurities carried forward by this step. The clearance credit belongs to PCR-005, PCR-007 and PCR-008.

| CQA                     | Crit. | Spec  | Acceptance | Mean $\pm$ SD  | Cpk   |
|-------------------------|-------|-------|------------|----------------|-------|
| Host Cell Protein (HCP) | M-H   | upper | 0-100      | 15.2 $\pm$ 4.6 | 6.143 |
| Residual DNA            | VL    | upper | 0-0.001    | 0.0001 $\pm$ 0 | 10.81 |

Host cell protein reaches a mean of 15.2 ng/mg against a limit of 100 ng/mg, a capability of 6.14, and residual DNA is capable by a wider margin at 10.81. Neither margin is earned here, and this report claims none of it. What this step contributes is a feed of consistent solids content and consistent impurity burden (turbidity and HCP), which is the condition under which the capture step performs as characterized in PCR-005.

The robustness of the step is expressed in its own terms. Step yield varies with a coefficient of variation of 2.0%, and clarification held within its normal operating range on every run but one over the loading and centrifugation ranges studied. That robustness statement is bounded by the normal operating ranges in Table 4.1, by the scale-down model of §3.1, and by the analytical precision given in Appendix B.

## 9 Parameter classification

Each parameter was classified under SOP-4001 from the demonstrated effects and from the risk of operating outside its range. The classifications are given in Table 4.1, and the reason for each is as follows.

- Centrifugation force is a key process parameter. It governs solids removal and therefore the load presented to the depth filter, and no effect on any quality attribute was seen across 6000–12000 g.
- Depth-filter loading is a key process parameter. It governs filter capacity and post-clarification turbidity, and it is readily controlled by fixing the filter area to the batch volume, so the risk of leaving its range is low.
- Post-clarification turbidity is a general process parameter. It is an outcome of the two settings above, it has no demonstrated link to a quality attribute, and it is monitored as the acceptance measure for the feed to capture.

The step carries no critical process parameter and no well-controlled critical process parameter, and of the 21 quality-linked parameters in the drug-substance control strategy none belongs to this step. The absence of an effect is not a gap in the study, and it is the evidence, gathered over ranges two to three times the width of the routine control ranges, that supports the classification above.

## 10 Contribution to the control strategy

The step contributes four elements to the control strategy of the drug substance. First, centrifugation force and depth-filter loading are held within the normal operating ranges of Table 4.1, which assures recovery and clarification. Second, post-clarification turbidity is monitored in-process on every batch (AMV-3015), with the upper limit of its normal operating range as the alert level and the upper edge of the characterized range as the action level. Third, depth filter media are flushed and integrity-tested before use under SOP-2006, with the flush volume set by the deviation described in §13.1. Fourth, the clarified harvest is accepted for capture against turbidity, which is the feed-consistency link to PCR-005.

Three things the step does not do should be equally clear. It clears no host cell protein and no DNA, so no clearance credit is taken for it anywhere in the package. It takes no viral-clearance credit, and the cumulative claim rests on PCR-006, PCR-008 and PCR-009. It forms no quality attribute, so it imposes no constraint on the design space of the drug substance. The contributions above roll up into the overall control strategy in PCMR-001.

# 11 Discussion

The step is well understood, and the understanding is simpler than that of any other step in the train. Its parameters act on solids and on recovery. They do not act on the antibody, and the study found nothing that qualifies that statement over ranges two to three times the width of the routine control ranges. The practical consequence is that the control strategy for this step is a process-performance control strategy, and no parameter of it needs the control that a critical process parameter would require.

The scale-down model is fit for the claims made from it. It reproduces the sedimentation duty of the commercial centrifuge and the loading duty of the commercial filter train, and qualification found no difference in yield or turbidity at the target condition. It does not reproduce the full shear history of the commercial machine and it operates at shorter hold times, but neither limit undermines the conclusions, because those conclusions concern recovery and solids removal and not a shear-sensitive attribute. Both are closed at scale during process performance qualification (PPQ).

Two deviations were recorded, and neither changed a conclusion. The more consequential of the two was a turbidity excursion to 12.5 NTU on a run loaded near the top of the characterized range. It was retained because the value stayed inside the characterized range and the root cause is the known capacity mechanism. The other deviation was a pre-use flush turbidity above its alert level, which was resolved by extending the flush before the filter was used. The details of both are in §13.

Three limitations should be carried forward. The univariate design cannot resolve an interaction between centrifugation and filter loading, which is accepted because neither acts on quality (§4.2). The characterization used harvest material from a single culture condition, so a change to the culture that materially altered the solids load would warrant a reassessment of filter capacity. Turbidity is an indirect measure of solids, and it does not distinguish a small quantity of large particles from a large quantity of fine ones.

## 12 Conclusions

Harvest and clarification is characterized and ready for Stage 2. The step forms no critical quality attribute and changes none, and its parameters were shown to have no effect on product quality over ranges two to three times the width of the routine control ranges. It recovers 97.7% of the product presented to it and delivers a clarified harvest at 5 NTU at the set-point condition.

Centrifugation force and depth-filter loading are key process parameters and post-clarification turbidity is a general process parameter, so the step carries no critical process parameter and it contributes no design space. The proven acceptable ranges of §7 are the characterized ranges, and they are bounded by the scale-down model and by the single culture condition studied. The impurity burden the step hands forward is cleared downstream, where host cell protein reaches a capability of 6.14 against its limit. Two deviations were recorded, and both were retained without effect on the classifications or the ranges. The outcome rolls up into PCMR-001.

## 13 Deviations from the plan

2 deviations from PCP-004 were recorded during execution, and each was investigated, root-caused and impact-assessed before the study was closed. Neither altered a parameter classification, a characterized range or a conclusion of this report. The register is given in Table 13.1, and each deviation is described below.

Table 13.1: Register of deviations recorded during execution of PCP-004.

| Devia-<br>tion | Summary                                                     | Detected during                 | Disposi-<br>tion |
|----------------|-------------------------------------------------------------|---------------------------------|------------------|
| DEV-004-01     | Depth-filter lot pre-use flush turbidity above alert        | pre-use filter flush            | retained         |
| DEV-004-02     | Post-clarification turbidity excursion above NOR on one run | in-process turbidity monitoring | retained         |

### 13.1 DEV-004-01 — pre-use flush turbidity above alert

The pre-use flush of a depth filter lot gave a turbidity of 6 NTU (AMV-3015), above the alert level applied to flush water, and the lot was LOT-DF-4408, allocated to clarification run 3. A filter that sheds media fines into the flush could carry those fines into the clarified harvest, which is why the alert exists.

The investigation compared the flush record with the volume specified in SOP-2006 and found that the initial flush volume applied was below it, and extending the flush brought the turbidity down, so that the filter was used only after the flush met its specification. The root cause is an insufficient initial flush volume and not a defect in the media, and the lot was released for use and the run was retained. The flush volume specified in SOP-2006 has been made an explicit step of the batch record, which is the control now carried into the control strategy (§10).

### 13.2 DEV-004-02 — post-clarification turbidity excursion

Post-clarification turbidity on one run reached 12.5 NTU, which is 2.5 NTU above the upper limit of the normal operating range of 1–10 NTU. The excursion was detected by in-process turbidity monitoring under AMV-3015 during the run, and the clarified harvest of that run was quarantined pending assessment.

The investigation established that the run was loaded to near the maximum of the characterized depth-filter range, at which capacity is nearly consumed, and that is the capacity mechanism described in §2.1, with the excursion its expected consequence at the edge of the range. The value stayed inside the characterized turbidity range of 1–20 NTU, and it is well outside the 4 % precision of the method, so it is a real effect and not assay noise. Product quality on that run was within the same variability as the rest of the study.

The run was retained and its data were used, and the disposition rests on two findings. The excursion occurred outside the normal operating range of depth-filter loading, so it does not describe routine operation, and the material remained acceptable for capture. The control that follows is the upper limit of 140 L/m<sup>2</sup> on loading, which is already the normal operating limit and is now supported by direct evidence of what happens above it.

# Appendices

## Appendix A — Product yield across the train

The step yield of harvest and clarification in the context of the whole drug-substance train, from the nominal commercial-scale simulation.

Table 13.2: Step and cumulative product yield across the drug-substance train.

| Step | Unit operation                                | Step yield | Cumulative yield |
|------|-----------------------------------------------|------------|------------------|
| 3    | Production Bioreactor                         | 1.0000     | 1.0000           |
| 4    | Harvest / Clarification                       | 0.9768     | 0.9768           |
| 5    | Protein A Chromatography                      | 0.9723     | 0.9496           |
| 6    | Low-pH Viral Inactivation                     | 0.9900     | 0.9401           |
| 7    | Cation Exchange Chromatography                | 0.9524     | 0.8954           |
| 8    | Anion Exchange Chromatography                 | 0.9728     | 0.8710           |
| 9    | Small Virus Retentive Filtration              | 0.9842     | 0.8573           |
| 10   | Ultrafiltration / Diafiltration (formulation) | 0.9707     | 0.8322           |

## Appendix B — Analytical methods summary

Performance of the validated methods applied at this step. Precision is the reported repeatability of the method and the limit of quantitation is given in the reportable unit. Entries marked as not recorded are methods whose performance summary is held in the method validation itself.

Table 13.3: Validated analytical methods applied to the harvest and clarification step.

| Reference | Method                          | Precision (%) | LOQ          | LOQ unit     |
|-----------|---------------------------------|---------------|--------------|--------------|
| AMV-3015  | Turbidity by Nephelometry (NTU) | 4.0           | 0.5          | NTU          |
| AMV-3012  | Host-Cell Protein ELISA         | 9.5           | 1.0          | ng/mg        |
| AMV-3014  | Residual DNA (qPCR)             | not recorded  | not recorded | not recorded |
| AMV-3011  | Size-Variants (SEC-HPLC)        | not recorded  | not recorded | not recorded |

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